

DIPLOMA DEFENSE · MAY 2026

Qonaqzhai

AI-powered event services marketplace for Kazakhstan.

Connecting customers, vendors, and traditions.

BAHAIDAHAR · ASTANA IT UNIVERSITY

01

The Problem

Event planning in Kazakhstan is fragmented, untrustworthy, and ignores local culture.

Status quo is broken

HOW IT WORKS TODAY

- Discovery via Instagram DMs
- WhatsApp chains for quotes
- No verified reviews
- Deposits to private cards = trust risk
- Customer juggles 5–10 vendors
- No platform supports Kazakh first-class
- Global tools ignore traditions

MARKET SIZE

\$1.5_B

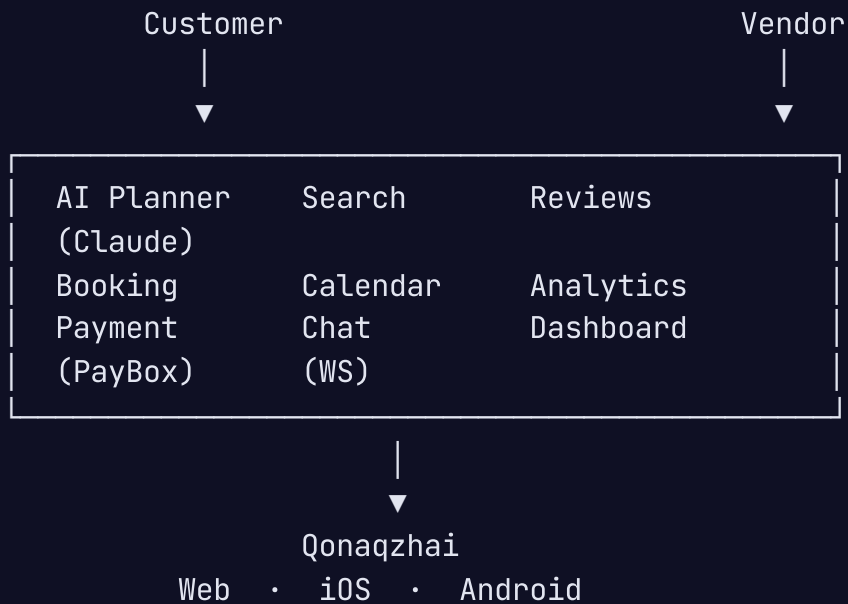
KZ events sector per year — fragmented across Instagram, fl.kz, weddingsalon.kz. **No platform owns the full journey.**

02

The Solution

Three clients, one platform. AI-first. Bilingual KZ / RU + EN.

Qonaqzhai at a glance



Trilingual KZ / RU / EN

AI event planner

PayBox · KZT-native

By the numbers

5

GO SERVICES

4

POSTGRES DBS

16

GRPC METHODS

68

HTTP ENDPOINTS

16

DB TABLES

13

MOBILE FEATURES

16

WEB ROUTES

2,033

GO TEST LOC

12

E2E SPECS

3

LANGUAGES

03

The Stack

Picked layer by layer against measured 2025-2026 industry data.

Technology decisions

BACKEND

Go 1.24

gRPC

Postgres 14

5 microservices · gateway / auth / core / payment / realtime

Protobuf contracts inter-service · HTTP/JSON at edge

JWT · PayBox PSP

WEB

Next.js 16

React 19

FSD

App Router · Cache Components · Partial pre-rendering

Tailwind v4 · Playwright E2E

MOBILE

Flutter 3.24

Riverpod

MVVM

Impeller engine · go_router · Dio + auth interceptor

Firebase Messaging · Secure storage

AI LAYER

Claude Sonnet 4.6

MCP

Tool-use event planner

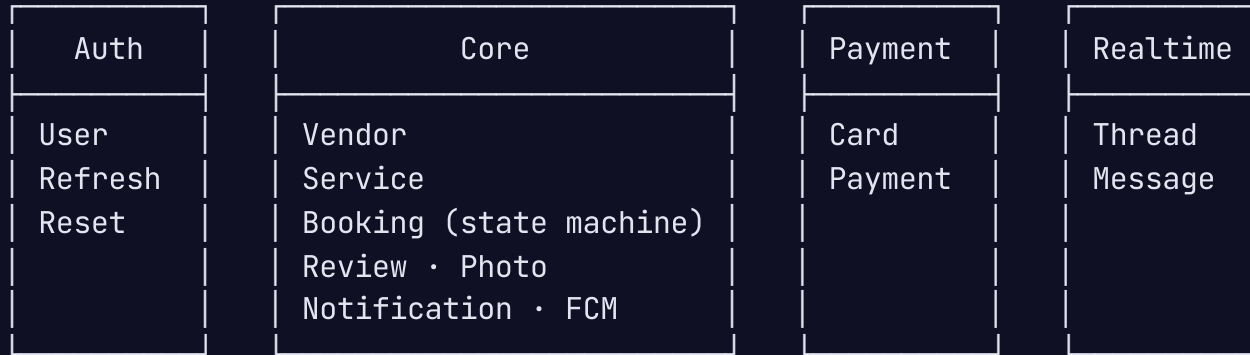
Agentic dev workflow with MCP servers

04

Architecture

Domain-driven · service-isolated · contract-first.

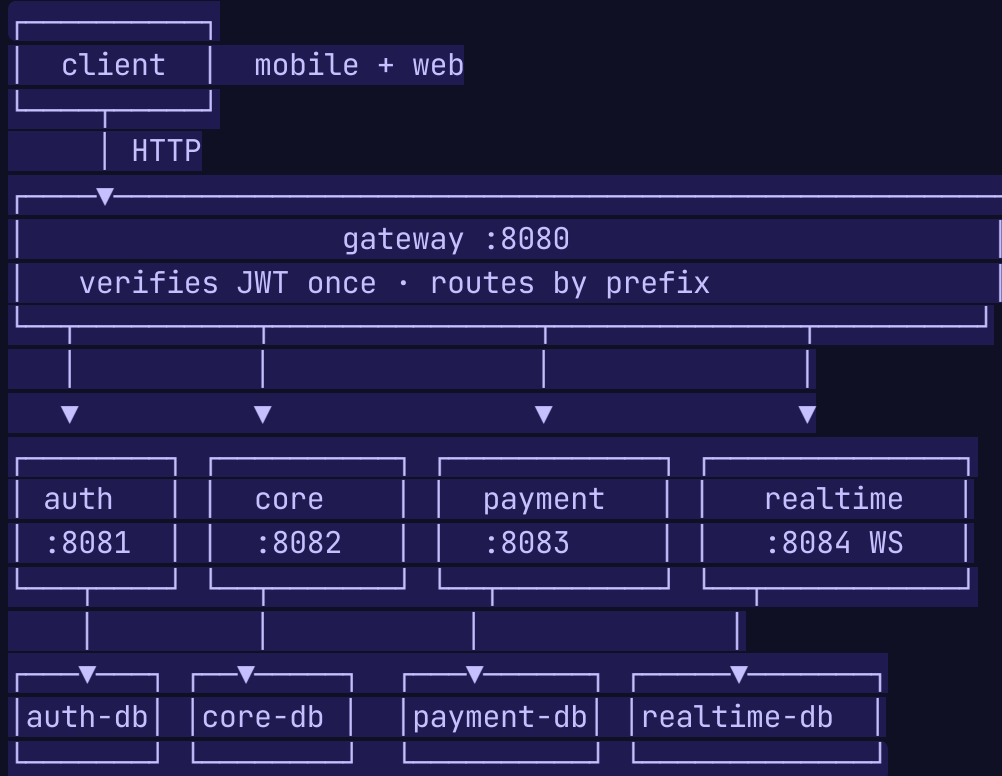
Domain model — four bounded contexts



Booking lifecycle — pending → accepted → completed → paid

Branches — declined · cancelled . Explicit state machine in core/usecase/booking/ .

Backend topology



One DB per service · gRPC inter-service · HTTP at edge

Service contracts — 16 gRPC methods

```
core —> auth      GetUser · GetUsersBatch · VerifyToken
core —> payment    Charge · Refund
core —> realtime   EnsureThread · PublishEvent
payment —> core    MarkBookingPaid          (saga callback)
realtime —> auth   GetUsersBatch           (peer-name enrich)
gateway —> auth    VerifyToken              (once per request)
```

PAYMENT SAGA — SYNCHRONOUS

```
core.booking.Pay → payment.Charge → core.MarkBookingPaid
```

Diploma scale — synchronous wins on simplicity.

Production migration path — event bus (NATS / Kafka).

Clean Architecture per service

```
backend/services/core/internal/  
├─ domain/      Pure types – Vendor, Booking, Review (no I/O)  
├─ ports/       Interfaces – VendorRepo, PaymentClient, Notifier  
├─ usecase/     Business logic – vendor/, booking/, review/, photo/  
└─ adapter/  
    ├─ http/     chi router · JSON marshalling · middleware  
    ├─ grpc/     gRPC server impl (core.proto)  
    ├─ grpcclient/ Outbound to auth, payment, realtime  
    ├─ repo/     PostgreSQL (sqlx)  
    └─ push/     FCM HTTP v1 sender
```

Layers depend inward only — `adapter → usecase → domain`. **Dependency Inversion** keeps usecase pure and testable. Table-driven Go tests stub the ports.

Web — Feature-Sliced Design

```
frontend/src/  
├── app/           Routes – vendors, bookings,  
                  threads, admin  
├── features/      ai-chat · auth · booking-flow  
                  messaging · reviews · services  
                  theme · vendor-search  
├── entities/      Shared domain types  
├── widgets/       Composed UI blocks  
└── shared/        UI kit · API · i18n · hooks
```

WHY FSD

Strict horizontal isolation. Dependency rule:

`shared → entities → features → widgets → app`

NEXT.JS 16 FEATURES USED

- Cache Components with "use cache"
- Partial pre-rendering on vendor lists
- React Server Components on vendor detail

Mobile — Flutter · Riverpod · Clean Arch

```
mobile/lib/  
├─ core/      network · router · theme · i18n · DI  
├─ features/  
│   13 modules — auth · vendor_catalog · booking  
│   ai_chat · payment · cards · reviews · messaging  
│   notifications · vendor_self · admin · onboarding  
│   settings  
└─ <feature>/  
    ├─ data/      DTOs · datasources · repository impl  
    ├─ domain/    Entities · abstract repo · use cases  
    └─ presentation/ ViewModels (Riverpod) · screens
```

Riverpod — compile-safe DI

go_router — deep links

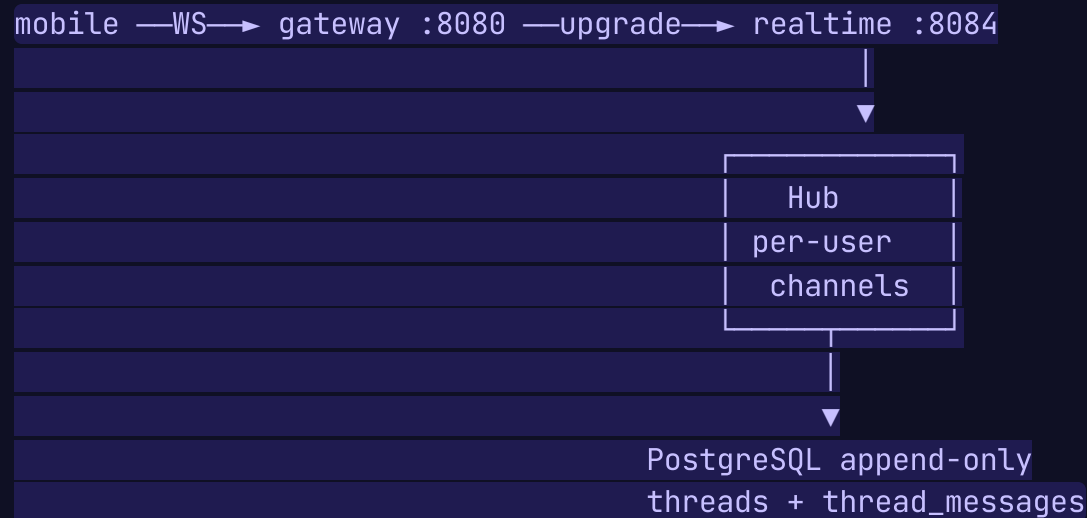
Dio — auto JWT refresh

Data model — 16 tables, 4 databases

DATABASE	TABLES	HIGHLIGHTS
auth-db	users , refresh_tokens , password_reset_tokens	role enum · hashed tokens
core-db	vendors , services , photos , bookings , reviews , notifications , fcm_tokens	full-text GIN · rating aggregation · BYTEA blobs
payment-db	cards , payments	last4 + brand · PAN never persisted
realtime-db	threads , thread_messages	one thread per booking · append-only

No cross-service foreign keys. `core.bookings.customer_id` is a plain UUID — `auth-svc` owns the actual `users` row. Cross-refs resolved via batched gRPC `auth.GetUsersBatch`.

Real-time chat — WebSocket hub



One thread per booking · Hub fans out and persists · JWT verified at upgrade · FCM deep-link fallback when peer offline

05

Why This Stack

Every choice backed by 2025–2026 industry data.

Stack justification — comparative

LAYER	CHOICE	ALTERNATIVE	WHY WE PICKED IT
Backend	Go 1.24	Node.js · Java	~2.6× CPU-bound · mature gRPC tooling
Inter-svc	gRPC + Protobuf	REST / JSON	5–10× throughput · 70–90% smaller payloads
Web	Next.js 16 + React 19	Remix · SvelteKit	Cache Components + PPR stable in v16
Web arch	Feature-Sliced Design	Atomic · flat	Hard horizontal isolation
Mobile	Flutter 3.24 Impeller	React Native	46% vs 35% share · 60–120 fps
Mobile state	Riverpod	Bloc · Provider	Compile-safe DI · no BuildContext
Database	Postgres per service	Shared DB	Service autonomy · independent scaling
AI	Claude API + MCP	OpenAI raw	MCP is cross-vendor standard (Linux Foundation)
E2E	Playwright	Cypress	Parallel cross-browser · TS-native

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Competitive Landscape

Where Qonaqzhai sits among global · CIS · Kazakh players.

Direct competitor matrix

PLATFORM	REGION	MODEL	AI	MOBILE	KZ TRADITIONS
The Bash	US	Commission	No	Yes	No
GigSalad	US	Freemium	Limited	Yes	No
Peerspace	US	Commission + escrow	Recs	Yes	No
The Knot	US	Listing + ads	Checklist	Yes	No
Wedding.ru	RU	Paid listings	No	Web - first	No
Eventie.kz	KZ	Paid listings	No	Web	Partial
weddingsalon.kz	KZ	Directory	No	Web	Partial
Qonaqzhai	KZ	Commission + escrow	Tool - use planner	Native Flutter	Yes

What competitors miss — what we built

GAPS WE FILL

- Kazakh-first UX (KZ / RU / EN equal)
- Local PayBox · KZT-native
- AI planner with real **tool use**
- Mobile-native, not web-first
- Tradition bundles — тұсауқесер · беташар · шашу

INSPIRATION PORTED IN

- Peerspace visual-first listings
- GigSalad freemium tier for vendors
- The Knot budget tracker
- Eventbrite promoted slots
- Peerspace escrow with milestone release

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Signature Feature

AI Event Planner with tool use.

Claude calls real backend tools

User: "Plan corporate event for 50 in Astana, 20 Aug, budget 1.5M ₸"

Claude Sonnet 4.6 (system prompt + tools)

search_vendors (core gRPC) check_availability (core gRPC) draft_booking (core gRPC) estimate_total (local calc)

Structured plan in chat

- Venue × 1 180 k ₸
- Catering × 50 900 k ₸
- Photographer × 1 200 k ₸
- Music DJ × 1 150 k ₸

Total: 1.43 M ₸ [Book all] [Customize]

Anthropic Messages API · `tools` parameter · JSON schemas · SSE streaming from `core` service.

08

Security & Quality

Defense in depth · table-driven tests · E2E gate.

Security model — defense in depth

LAYER	MECHANISM
Transport	HTTPS at gateway · Let's Encrypt
AuthN	JWT HS256 · 15 min access + 7 day refresh · hashed
AuthZ	RBAC middleware · per-route guards · ownership checks
Edge	Gateway calls <code>auth.VerifyToken</code> on every request
Service	Each service re-verifies JWT — no implicit trust
Rate limit	Per-IP 20–40 req / 10 s on every service
Password reset	One-time short-TTL tokens · SMTP delivery
PCI scope	PAN never persisted — PayBox PSP holds the scope
Photo uploads	MIME whitelist · 5 MB cap · content-type sniffed
Cross-DB joins	Forbidden — no shared schema, no leaked FKs

Testing & quality

COMPONENT	FILES	LOC	APPROACH
Auth service	1	386	Signup · login · refresh · password reset
Core service	4	1,007	Vendor CRUD · booking state machine · review aggregation
Payment service	2	436	Card validation · charge · refund · PSP mock
Realtime service	1	204	Thread ensure · message send · WS connection
Backend total	8	2,033	Table-driven Go tests · <code>-race</code> enabled
Frontend E2E	12	~800	Playwright — auth · booking · chat · admin · vendor
Mobile	smoke	growing	Riverpod <code>ProviderContainer</code> mocks

CI gate — backend tests + Playwright E2E must pass before merge.

09

Agentic Engineering

MCP, Claude, and how this diploma was actually built.

How this diploma was built

MCP SERVERS USED IN DEVELOPMENT

- **GitHub MCP** — branch / PR / issue automation
- **Filesystem MCP** — repo-wide refactors
- **Postgres MCP** — schema introspection
- **Context7 MCP** — Next.js 16 + Riverpod docs
- **Exa / WebSearch** — competitor research

WHY THIS MATTERS IN 2026

- MCP donated to **Linux Foundation Agentic AI Foundation** (2025)
- **10K+** MCP servers live · **97M+** SDK installs
- Stack Overflow 2025 — **84%** of developers use AI tools
- IDC — **\$36.2B** AI dev-tool market by end-2026

This diploma mirrors how engineering is actually done today.

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Demo & Roadmap

What you'll see today, and what ships next.

Live demo — 8 minutes

- 1 **Localization** · 30s · Open mobile · switch KZ → RU → EN
- 2 **AI Planner** · 2 min · "Plan corporate event for 50 in Astana, 20 Aug, 1.5M ₸" · Claude returns 4-vendor plan with reasoning
- 3 **Booking + Calendar** · 1 min · Tap venue · view availability · confirm Aug 20
- 4 **Payment** · 1 min · PayBox test card · push notification · deep-link into booking
- 5 **Vendor side** · 1 min · Log in as vendor · see new booking in analytics · revenue chart updates
- 6 **Admin moderation** · 30s · Web admin moderates flagged review
- 7 **Architecture trace** · 1 min · `docker compose logs` — saga `core → payment.Charge → core.MarkBookingPaid`
- 8 **Q & A** · 1 min

Roadmap

TIER 1 — SHIPS BEFORE DEFENSE

- AI event planner with tool use
- Vendor analytics dashboard
- KZ / RU / EN i18n completion
- Push notifications with deep-links
- Chat polish — typing · read receipts

TIER 2 — NEXT ITERATION

- Smart vendor recommendations
- Dynamic pricing (peak / off-peak)
- Google Calendar sync
- Referral · promo codes
- Verified-booking review badge

TIER 3 — RESEARCH

- Escrow with milestone release
- Voice input for AI planner
- ML matching via embeddings
- KYC via egov.kz public API
- AI-generated event runbook

Lessons learned — honest retrospective

WHAT WORKED	WHAT I WOULD CHANGE	
One DB per service — clear ownership	Synchronous payment saga is fragile · publish events instead (NATS / Kafka)	
Riverpod for Flutter — fewer rebuilds	Hand-rolled DTOs · should generate from OpenAPI from day 1	
FSD on web — features stay isolated	Tailwind v4 + Next.js 16 still rough · hydration warnings remain	
Claude + MCP — 2–3× productivity	Need stricter prompt discipline · early sessions drifted scope	
Per-service Go modules — independent CI	Workspace <code>replace</code> directives are subtle · broke CI twice	
gRPC for internal — no API drift	Browser debugging painful · we keep REST at the gateway for a reason	

THANK YOU

Questions?

Repo · github.com/Bahaidahar/diploma

Architecture deep-dive · [presentation/research/code-map.md](#)

Stack justification · [presentation/research/stack-justification.md](#)

Comparative analysis · [presentation/research/competitors.md](#)